

JPMorgan Chase & Co.

IdxAi.

AI-Powered Thematic Investment Intelligence

GPT-4 · Generative AI · Thematic Indexing · Agentic Finance

6–9 Months

→ Few Hours

\$M Cost

→ Near Zero

300+

Thematic Baskets

GPT-4

Core Engine

EXECUTIVE SUMMARY

What is IndexGPT?

IndexGPT is JP Morgan Chase's generative AI-powered thematic investment platform, unveiled in 2023–24 to automate the creation of investment baskets based on emerging financial trends.

Key Highlights:

- Core Engine: OpenAI GPT-4 + specialized NLP model
- Trademark filed: May 2023 (USPTO)
- Product type: Agentic, multi-stage AI pipeline
- Target: Institutional → Retail investors
- First major bank to release GPT product to customers

Business Impact

Speed

6–9 months → hours

Cost

Millions → Near Zero

Accuracy

2× more keywords vs legacy tools

Real-Time

Always-on, autonomous rebalancing

Coverage

Equities, Bonds, ESG, Crypto

PROBLEM STATEMENT

Why JPMorgan needed to reinvent thematic investing

01

Months-Long Index Creation

Building a single thematic investment basket required 6–9 months of manual research by teams of analysts — making niche, fast-moving themes commercially unviable.

02

Million-Dollar Cost Per Index

Legacy processes cost millions of dollars per index, limiting portfolio customization only to large institutional mandates and blocking retail democratization.

03

Static, Backwards-Looking Portfolios

Traditional sector indices lagged real-world trends. By the time an 'e-sports' or 'cloud computing' index was ready, the market opportunity had already partially played out.

04

No Autonomous Analyst Coverage

Human analysts couldn't continuously screen thousands of companies across news, SEC filings, and earnings — creating blind spots in theme tracking and security selection.

SOLUTION OVERVIEW

IndexGPT: A Multi-Stage Agentic AI Pipeline for Thematic Investing

Robust & Scalable Architecture

- Primary AI Core: OpenAI GPT-4 for theme generation & keyword extraction
- Secondary Engine: Specialized NLP model for high-volume news & sentiment scanning
- Architecture Style: Agentic AI — autonomous task execution without human-in-the-loop
- Infrastructure: Cloud-native, elastic compute for real-time market data processing
- Data Sources: Market data, news articles, SEC filings, economic indicators, corporate reports
- Asset Coverage: Equities, Fixed Income, ESG, Cryptocurrencies

Thematic Indexing

Niche trend-based baskets (e.g. 'Remote Work' not just 'Tech')

Personalization

Tailored portfolios by risk tolerance, goals & preferred themes

Real-Time Adaptation

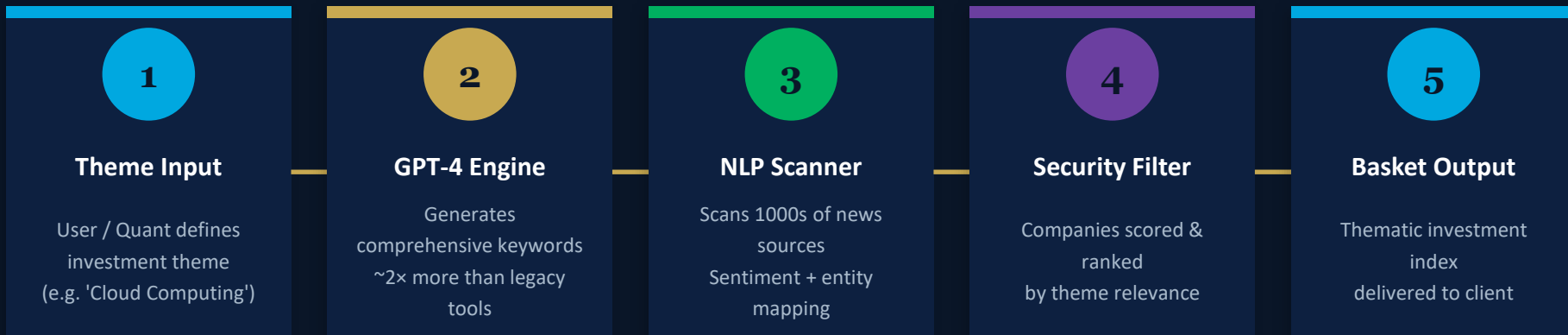
Continuous SEC filing & news screening — always-on analyst

Democratization

Reduces cost to near zero, enabling retail investor access

TECHNICAL ARCHITECTURE

Multi-Stage Agentic Pipeline — From Theme to Investment Basket



GPT-4 (OpenAI)

Theme gen & keyword extraction. Produces 2× keyword density vs prior software.

Specialized NLP

Secondary model for high-volume article scanning & entity recognition at scale.

Cloud Infrastructure

Elastic compute handles real-time market data; deployed as cloud-native SaaS.

Agentic Autonomy

System autonomously reads filings, screens companies — a virtual analyst team.

TECHNICAL WORKFLOW — INTERNAL DETAILS

Step 1: Theme Generation & Keyword Extraction

- GPT-4 receives a high-level theme (e.g. 'E-Sports', 'Genomics')
- Generates a comprehensive keyword taxonomy covering all sub-sectors
- Produces ~2× more relevant keywords than prior software solutions
- Keywords structured as a semantic graph for downstream NLP processing

Step 2: News & Sentiment Scanning

- Keywords fed into the secondary NLP model
- Model scans thousands of news articles, press releases & media sources
- Identifies companies actively mentioned in context of the theme
- Sentiment scoring applied: positive/negative association weighted

Step 3: Security Selection & Basket Construction

- Companies filtered by relevance score, market cap & liquidity criteria
- Final basket of securities constructed — the 'thematic investment index'
- Supports equities, fixed income, ESG-focused assets & cryptocurrencies
- Output delivered as a structured index to portfolio management systems

Step 4: Real-Time Monitoring & Rebalancing

- System acts as 'always-on' autonomous analyst post-basket creation
- Continuously re-screens financial reports, news & earnings data
- Triggers rebalancing signals when company relevance scores shift
- No human-in-the-loop required for ongoing monitoring

AI MODELS & HIDDEN TECHNICAL DETAILS

Under the hood: What powers IndexGPT



Primary Model: OpenAI GPT-4

- Model Type: Large Language Model (Generative Pre-trained Transformer)
- Role: Theme understanding, keyword taxonomy generation, semantic reasoning
- Key Capability: Contextual understanding of financial themes across macro & micro domains
- Output: Structured keyword lists + semantic relationships between entities
- Advantage: Produces 2× keyword depth compared to traditional keyword tools
- JP Morgan's filing: explicitly references 'Generative Pre-trained Transformer (GPT) models'



Secondary Model: Specialized NLP

- Model Type: Purpose-built Natural Language Processing model
- Role: High-volume text processing AFTER GPT-4 generates keywords
- Input: Keyword taxonomy from GPT-4 + corpus of news/financial documents
- Processing: Entity recognition, sentiment analysis, company mention scoring
- Scale: Scans thousands of articles & filings per theme in minutes
- Why separate NLP? GPT-4 excels at reasoning; specialized NLP is faster/cheaper for bulk scan

DATA SOURCES, COVERAGE & SPECIFICATIONS

Data Inputs



Market Performance Data

Live price feeds, volume, market cap, momentum signals across global exchanges



News Articles & Media

Real-time scan of financial news, press releases, earnings transcripts & analyst reports



Economic Indicators

Macro signals: GDP, inflation, sector rotation triggers, central bank data



Corporate Financial Reports

SEC filings (10-K, 10-Q, 8-K), EDGAR data, international equivalents

Asset Classes & Specs

Equities

Global stocks, growth & value, sector & sub-sector

Fixed Income

Government & corporate bonds, duration strategies

ESG Assets

Environment, Social & Governance screened baskets

Cryptocurrencies

Digital assets integrated as thematic exposure

Technical Specifications

Trademark: USPTO — 'cloud computing software using AI for analyzing and selecting securities tailored to customer needs' | Deployment: Cloud-based SaaS
| Model: GPT architecture (OpenAI GPT-4)

BUSINESS VALUE ACHIEVED

6–9 Mo

→ Few Hours

Index Creation Time

~\$0

vs. Millions

Cost Per Index

2×

Keyword Depth

vs Legacy Tools

24/7

Autonomous

Market Monitoring

Operational Efficiency

- Index creation compressed from months to hours — a fundamental step-change in speed-to-market
- Analysts freed from manual keyword research to focus on strategy and client advisory

Cost Optimization

- Index creation cost reduced from millions to near zero per thematic basket
- Enables profitability on hyper-niche indices that were previously commercially unviable

Scalability & Market Position

- First major bank to release a GPT-like product directly to customers (per USPTO filing + attorney)
- Positions JPMorgan as lead innovator in AI-driven financial services globally

Business Growth

- Institutional clients: superior theme representation; Retail roadmap: Latin America pilot
- Continuous real-time rebalancing replaces periodic manual reviews — living portfolios

STRATEGIC CONCLUSION

IndexGPT — Redefining What's Possible in Financial AI

JP Morgan's IndexGPT represents the most ambitious application of Generative AI in mainstream financial services — combining GPT-4's reasoning with specialized NLP at scale to autonomously construct, monitor, and rebalance thematic investment portfolios. It does not merely automate existing processes; it makes entirely new product categories commercially viable.

First Mover Advantage

First bank to patent & launch a GPT product to customers — 3-year trademark window creates strong moat

Democratization of Alpha

Ultra-niche themes (e.g. 'AI in Agriculture') now viable for retail — unlocking a new revenue category

Autonomous Finance

IndexGPT is a blueprint for agentic AI in finance — the beginning of AI-run portfolio management

Key Facts Recap

Trademark: IndexGPT (USPTO, May 2023) · Models: GPT-4 + Specialized NLP · Pipeline: 4-Stage Agentic AI · Cost: Near Zero per index · Time: Hours vs Months · Audiences: Institutional + Retail · Coverage: Equities, Bonds, ESG, Crypto